

PI-LNO: Physics-Informed Surrogates for Heterogeneous-Agent Models

Brandon Yee^{a,*}, Anqi Peter Li^a, Pairie Koh^{a,b}

^a*Yee Collins Research Group*

^b*Free Systems Lab, Stanford University*

Abstract

Simulation-based optimization of complex stochastic systems is bottlenecked by the cost of repeatedly solving the underlying model. We propose PI-LNO, a physics-informed neural operator that serves as a drop-in AI surrogate for a coupled PDE system arising in heterogeneous-agent economic models, replacing an iterative finite-difference solver with a single neural network forward pass. The architecture combines a Fourier neural operator (FNO) backbone with FiLM parameter conditioning and three structural constraints that preserve the mathematical properties of the original system: density normalization (valid probability measure), first-order-condition-derived policy (value-policy consistency), and adjoint-consistent physics losses matching the classical solver’s discretization. A staged curriculum introduces physics residuals progressively, preventing destabilizing feedback during training. PI-LNO produces value functions, policies, distributions, and equilibrium prices in a single forward pass, achieving a $26\times$ single-query and $7,590\times$ batched speedup over a CPU finite-difference solver on an NVIDIA L4 GPU (a GPU-accelerated solver would narrow this gap; the claim is end-to-end estimation feasibility). Ablations on the matched 3-stage pipeline (5-seed) show FiLM conditioning contributes $\sim 2.4\times$ on value error and $\sim 2\times$ on distributional moments; an MLP baseline (FNO blocks replaced by pointwise MLPs, single-stage training, 594K vs. 1,685K parameters; not matched to the 3-stage curriculum) is consistently weaker on every distributional metric, though the gap conflates architecture and training differences. The surrogate enables Simulated Method of Moments estimation and identification analysis in seconds; FD-solver controls at all three evaluation points show θ^* is a well-separated optimum (5 starts, 7D space), with the recovered parameter bias attributable to weak global identification amplifying small surrogate moment-surface error. The structural design principles (adjoint-consistent physics losses, FOC-derived policies, staged curriculum) transfer to other coupled PDE systems where structure-preserving surrogates can accelerate simulation-based inference; within the Aiyagari framework, extension to richer income processes ($J \geq 3$ states) requires re-implementing the physics losses for the new generator structure and is an open direction.

Keywords: neural operator, physics-informed learning, surrogate optimization, simulation-based estimation, heterogeneous-agent models, Fourier neural operator

*Corresponding author.

Email addresses: b.yee@ycrg-labs.org (Brandon Yee), p.li@ycrg-labs.org (Anqi Peter Li), p.koh@ycrg-labs.org (Pairie Koh)

1. Introduction

Simulation-based optimization of complex stochastic systems is a core challenge in operations research: the forward model must be solved repeatedly (once per candidate parameter vector) and the computational cost of each evaluation constrains the scope of feasible inference. This paper addresses this challenge for a class of coupled partial differential equation (PDE) systems that arise in economic equilibrium modeling, where a single solve requires three nested iterative loops and estimation demands $\mathcal{O}(10^3\text{--}10^4)$ such evaluations. We introduce a physics-informed neural operator surrogate that replaces the iterative solver with a single forward pass, reducing estimation time from hours to seconds. The central technical challenge (shared with surrogate-accelerated optimization more broadly) lies in ensuring that the surrogate preserves sufficient structural fidelity to yield reliable parameter estimates, not merely fast ones.

Our application domain is heterogeneous-agent (HA) economic models, the workhorse of modern quantitative macroeconomics. The Bewley–Huggett–Aiyagari framework [1, 2, 3] models a continuum of agents choosing consumption and saving under idiosyncratic income risk and incomplete markets (§2). In its continuous-time formulation [4], equilibrium requires jointly solving three components: a Hamilton–Jacobi–Bellman (HJB) equation governing optimal agent behavior, a Kolmogorov Forward equation (KFE) governing the stationary cross-sectional wealth distribution, and a market-clearing condition. The market-clearing condition endogenizes the interest rate, coupling all three components. These models are central to policy analysis (monetary policy transmission [5, 6] and inequality dynamics [7, 8]) but the computational cost of solving them creates a structural bottleneck that constrains both the scope of models amenable to estimation and the rigor of inference that can be conducted. The structural design principles (adjoint-consistent physics losses, FOC-derived policies, and staged curriculum training) transfer to other simulation-based optimization settings where the forward model involves multi-physics coupling, endogenous closure conditions, and boundary singularities. The current implementation is calibrated to the two-state ($J = 2$) Poisson income process; extension to $J \geq 3$ states requires re-deriving the adjoint-consistent discrete operators for the new generator structure and is left for future work.

The HJB and KFE are not independent PDEs: they are formal adjoints of each other, linked through the household’s optimal savings policy. This adjoint duality is the structural backbone of the equilibrium, and the classical solver exploits it explicitly by constructing the KFE operator as the transpose of the HJB’s drift matrix. We argue that a neural surrogate must preserve this duality to produce reliable downstream inference. Doing so requires deliberate design choices in the architecture, loss function, and training schedule, choices whose necessity we establish both analytically and through ablation. We detail each design decision and characterize the failure modes that arise when components are omitted through ablation experiments.

1.1. The Computational Bottleneck

Solving the continuous-time Aiyagari model on a 64×2 grid (wealth $\times$ income states) takes 0.1–0.5 seconds per parameter configuration using the implicit finite-difference scheme of [4]. The cost arises from three nested iterative loops. The inner loop iterates the HJB equation to convergence via implicit Euler with aggressive damping ($\Delta t = 1000$), requiring $\mathcal{O}(10^1\text{--}10^2)$ steps. The middle loop solves the KFE as a linear system given the converged policy. The outer loop performs bisection on the interest rate until market clearing is satisfied to tolerance. Each level depends on the output of the one below it, and the entire stack must be re-executed from scratch for every new parameter vector θ .

Structural estimation via simulated method of moments (SMM) [9] requires $\mathcal{O}(10^3\text{--}10^4)$ evaluations for derivative-free optimization. With 10^4 evaluations at 0.1–0.5 seconds each, estimation takes 15 minutes to over an hour on a single core. A substantial fraction of random parameter configurations fail to converge, necessitating up to $20\times$ oversampling to reach the target sample count. Extension to richer models (additional income states, aggregate shocks, multiple assets) compounds the cost multiplicatively. This bottleneck is structural rather than computational: each parameter configuration is treated as an independent fixed-point problem, with no information shared across evaluations.

1.2. Challenges Beyond Standard PDE Surrogates

Neural operators [10, 11] learn mappings between function spaces, amortizing the cost of parametric PDEs across the full parameter domain. However, applying neural operators to HA equilibrium models introduces three challenges absent in standard PDE surrogates:

1. **Forward-backward coupling.** The HJB equation propagates information backward (the value at any wealth level depends on future optimal behavior) while the KFE propagates information forward, as the distribution evolves under the optimal policy. Standard PDE surrogates approximate a single directional mapping; here the network must implicitly represent a fixed point of two coupled PDEs with opposing information flows. A surrogate that independently approximates V^* and μ^* will generically violate this coupling: the predicted distribution will not be stationary under the predicted policy.
2. **Borrowing constraint singularity.** At the borrowing limit $a = a_{\min}$, the value function develops a kink ($\partial_{aa}^2 V^* \rightarrow \infty$) as the consumption policy transitions from constrained ($c^* = wz + ra$, consuming all income) to unconstrained behavior. Simultaneously, the wealth distribution develops a mass concentration as constrained households accumulate at the boundary. These co-located singularities in both the backward and forward solutions imply that small approximation errors at the boundary propagate non-locally through the coupled system.
3. **Equilibrium closure.** The interest rate r is not an input but an endogenous equilibrium object determined by market clearing. The solution operator $\mathcal{G}(\theta)$ must internally resolve the aggregate consistency condition $K^s(r) = K^d(r)$, which depends on integrating the predicted

distribution. This makes the operator’s output globally coupled: changing the predicted density at any wealth level changes the implied interest rate, which changes every value function and policy.

1.3. Our Approach: Structure Preservation

Each of the three challenges above corresponds to a structural relationship in the equilibrium that a naive surrogate would break. Our PI-LNO preserves these relationships explicitly, at every level of the design:

- **Forward-backward coupling $\rightarrow$ adjoint-consistent loss.** We compute HJB and KFE residuals using the *same* upwind discretization, ensuring the discrete KFE operator is exactly the transpose of the discrete HJB operator. This is the mechanism by which the classical solver couples the two PDEs, and we replicate it in the loss function (Section 5). Adjoint consistency is automatic for any predicted drift $\hat{s}$; the practical question is whether $\text{sign}(\hat{s}) = \text{sign}(s^*)$ at each grid point so the correct upwind branch is selected. Across the 1,500-example test set, sign agreement is 99.1%, 99.3%, and 99.4% at Stages 1, 2, and 3 respectively (computed over all grid points with $|s^*| > 10^{-8}$), confirming the upwind branches are correct in practice throughout training.
- **Borrowing constraint singularity $\rightarrow$ constrained architecture.** Consumption is derived from value-function gradients via the FOC, not predicted independently, so the value-policy link cannot break. The density is output through a softplus activation with normalization that guarantees $\hat{\mu} \geq 0$ and $\int \hat{\mu} = 1$ (Section 4).
- **Equilibrium closure $\rightarrow$ staged curriculum.** The market-clearing residual couples the rate head to the density head globally. Introducing this loss prematurely (before the density is approximately correct) creates destabilizing feedback. A staged curriculum resolves this by allowing the data fit to stabilize before physics losses are activated (Section 5).

Consequently, PI-LNO is designed to produce outputs that are *structurally coherent*: the architecture and loss function encourage the predicted equilibrium to approximately satisfy the same coupled system of equations that defines the true equilibrium. The degree to which this coherence is achieved in practice depends on the training configuration and is evaluated empirically in Section 6.

Scope. We restrict attention throughout to the two-state Poisson income process ($J = 2$ with symmetric switching intensity λ). Extending the physics losses and training pipeline to multi-state income processes requires re-implementing the generator structure; we defer this generalization to future work.

1.4. Contributions

Organized around the structure-preservation principle, our contributions are:

1. **Problem analysis.** We identify three structural properties of HA equilibria (forward-backward coupling, constraint singularity, and equilibrium closure) that standard PDE surrogates violate, and show each demands a specific design response (§1.2).
2. **Structure-preserving architecture.** We design PI-LNO, a latent-conditioned FNO with FiLM modulation, whose constrained output heads enforce economic structure by construction: FOC-derived consumption and normalized softplus density (§4).
3. **Adjoint-consistent training.** We construct the discrete HJB and KFE residuals as exact transposes, introduce them via a staged ramp that respects the forward-backward coupling, and document the failure modes that arise when this structure is not preserved (§5).
4. **Estimation pipeline.** We embed PI-LNO in a simulation-based estimation pipeline (SMM) with SVD-based identification diagnostics, demonstrating that the surrogate’s speed enables dense exploration of the moment surface and reveals weak identification structure invisible to classical solvers (§7). An FD-solver control confirms that parameter-recovery bias reflects model-level identification weakness, not surrogate error.

2. The Continuous-Time Bewley–Huggett–Aiyagari Economy

We present the economic model in sufficient detail to formalize the adjoint relationship between the HJB and KFE equations, following [4].

2.1. Households

A unit continuum of infinitely-lived households, indexed by wealth $a \in [a_{\min}, \bar{a}]$ and idiosyncratic productivity $z \in \{z_1, z_2\}$, choose consumption to maximize expected discounted utility:

$$\max_{\{c_t \geq 0\}} \mathbb{E}_0 \int_0^\infty e^{-\rho t} u(c_t) dt, \quad u(c) = \begin{cases} \frac{c^{1-\sigma}-1}{1-\sigma} & \sigma \neq 1, \\ \log c & \sigma = 1, \end{cases} \quad (1)$$

subject to the flow budget constraint

$$\dot{a}_t = wz_t + ra_t - c_t, \quad a_t \geq a_{\min}, \quad (2)$$

where w is the wage, r is the interest rate, and $a_{\min} \geq 0$ is the lower bound on asset holdings. In our parameterization, $a_{\min} \in [0, 0.5]$ (Table 1), which functions as a *minimum savings requirement* rather than a borrowing limit in the conventional sense; agents cannot hold negative wealth. Productivity follows a continuous-time Markov chain with generator $\Lambda = \begin{bmatrix} -\lambda & \lambda \\ \lambda & -\lambda \end{bmatrix}$ and symmetric switching intensity $\lambda > 0$. The two income states are constructed symmetrically around the mean: $z_1 = \bar{z}(1 - \eta)$ and $z_2 = \bar{z}(1 + \eta)$, where $\eta = 0.30$ is a fixed income spread parameter.

2.2. Firms

A representative firm operates Cobb–Douglas technology $Y = AK^\alpha L^{1-\alpha}$ with total factor productivity $A > 0$ and $\alpha \in (0, 1)$. Competitive factor markets yield:

$$r = A\alpha K^{\alpha-1} L^{1-\alpha} - \delta, \quad w = A(1 - \alpha) \left(\frac{K}{L} \right)^\alpha, \quad (3)$$

where $\delta > 0$ is the depreciation rate, $A = 0.1$ is the TFP normalization, and $L = \sum_j z_j \pi_j$ is aggregate effective labor (with π the ergodic distribution of z ; equal to $\bar{z}$ in the symmetric two-state case).

2.3. Stationary Equilibrium

Definition 1 (Stationary Competitive Equilibrium). *A stationary competitive equilibrium consists of a value function $V : [a_{\min}, \bar{a}] \times \{z_1, z_2\} \rightarrow \mathbb{R}$, a consumption policy $c^* : [a_{\min}, \bar{a}] \times \{z_1, z_2\} \rightarrow \mathbb{R}_{>0}$, a wealth density $\mu : [a_{\min}, \bar{a}] \times \{z_1, z_2\} \rightarrow \mathbb{R}_{\geq 0}$, and an interest rate $r^* \in \mathbb{R}$ satisfying the three conditions (E1)–(E3) below.*

Section 4 shows how PI-LNO encodes these structural constraints (optimality, stationarity, and market clearing) directly into its output heads and loss construction. The normalization and FOC constraints are enforced exactly by construction; the HJB, KFE, and market-clearing constraints are soft-enforced via the loss function and hold approximately in practice (clearing residual $\approx 2.5\%$ of mean capital stock at Stage 3; Section 6.1).

(E1) Household optimality (HJB). The value function satisfies:

$$\rho V(a, z_j) = \max_{c > 0} [u(c) + \partial_a V(a, z_j) \cdot (wz_j + r^*a - c)] + \sum_{k=1}^2 \Lambda_{jk} V(a, z_k), \quad (4)$$

for all $(a, z_j) \in (a_{\min}, \bar{a}] \times \{z_1, z_2\}$, with the state constraint $\dot{a} \geq 0$ at $a = a_{\min}$ enforced via $\partial_a V(a_{\min}, z_j) \geq u'(wz_j + r^*a_{\min})$. The first-order condition yields:

$$c^*(a, z_j) = (\partial_a V(a, z_j))^{-1/\sigma}, \quad s^*(a, z_j) = wz_j + r^*a - c^*(a, z_j). \quad (5)$$

The borrowing constraint binds when $s^*(a_{\min}, z_j) \leq 0$, in which case $c^*(a_{\min}, z_j) = wz_j + r^*a_{\min}$.

(E2) Stationarity (KFE). The wealth density satisfies the Kolmogorov Forward equation:

$$0 = -\partial_a [s^*(a, z_j) \mu(a, z_j)] + \sum_{k=1}^2 \Lambda_{kj} \mu(a, z_k), \quad \forall (a, z_j), \quad (6)$$

with reflecting barrier at $a = a_{\min}$ (flux $s^*\mu$ vanishes at the boundary), normalization $\sum_j \int \mu(a, z_j) da = 1$, and $\mu \geq 0$.

The KFE is the adjoint of the infinitesimal generator of the household's state process. If A denotes the linear operator on the right-hand side of the HJB (after substituting the optimal policy), then the KFE operator is A^* , its L^2 -adjoint. This adjoint relationship is the structural backbone of

the equilibrium: the KFE operator is determined by, and must be consistent with, the HJB operator. Preserving this relationship at the discrete level is the central design constraint of our approach.

(E3) Market clearing. The interest rate r^* clears the capital market:

$$K^s(r^*) \equiv \sum_j \int_{a_{\min}}^{\bar{a}} a \mu(a, z_j) da = K^d(r^*) \equiv \left(\frac{A\alpha}{r^* + \delta} \right)^{1/\alpha} L. \quad (7)$$

2.4. The Fixed-Point Structure

The equilibrium has a nested fixed-point structure:

1. **Outer loop (interest rate):** Given r , the household problem and distribution are well-defined. The excess demand function $\Xi(r) = K^s(r) - K^d(r)$ is continuous and strictly decreasing, so r^* is its unique root.
2. **Middle loop (distribution):** Given r and the converged policy $c^*(\cdot; r)$, the KFE is a linear equation in μ , a single linear solve.
3. **Inner loop (value function):** Given r , the HJB is solved via implicit Euler: $(I + \Delta t A) V^{(n+1)} = V^{(n)} + \Delta t u(c^{(n)})$, converging when $\|V^{(n+1)} - V^{(n)}\|_\infty < 10^{-7}$ (up to 400 iterations).

The information flow is $r \rightarrow V^*(r) \rightarrow c^*(r) \rightarrow \mu^*(r) \rightarrow K^s(r) \rightarrow \text{update } r$. A neural operator must capture all three levels of this fixed point in a single forward pass.

2.5. Structural Parameter Space

The structural parameter vector is

$$\theta = (\rho, \sigma, a_{\min}, \lambda, \bar{z}, \alpha, \delta) \in \Theta \subset \mathbb{R}^7, \quad (8)$$

comprising the discount rate, CRRA curvature, borrowing constraint, income switching intensity, mean productivity, capital share, and depreciation. Training ranges are given in Table 1. Not every parameter combination in Θ admits a stationary equilibrium: when risk aversion and discounting are simultaneously high while income persistence is strong, the equilibrium interest rate exceeds the feasible ceiling $r < \rho$, and auto-bracketing fails. We generate training data by rejection sampling, drawing uniformly from Θ and retaining only draws for which the solver converges. At the default grid resolution ($n_a = 64$), approximately 69% of draws succeed; the effective training distribution is therefore a censored subset of the declared box, concentrated in the region where the equilibrium rate lies comfortably below ρ .

3. Related Work

Our work sits at the intersection of three literatures: neural operators for parametric PDEs, physics-informed machine learning, and surrogate-accelerated simulation-based inference. The specific combination of coupled forward-backward PDE structure with endogenous equilibrium

Table 1: Structural parameters, training bounds, and roles in the equilibrium system.

Symbol	Name	Lower	Upper	Where it enters
ρ	Discount rate	0.03	0.08	HJB: discounts future utility
σ	Risk aversion	1.0	4.0	FOC: $c = V_a^{-1/\sigma}$; log utility at $\sigma = 1$
$a_{\min}$	Min. asset holding	0.0	0.5	Grid boundary; constraint mass
λ	Income switching	0.05	0.30	Generator Λ
$\bar{z}$	Mean productivity	1.0	1.8	Budget: $wz + ra$
α	Capital share	0.28	0.38	$K^d = (\alpha/(r+\delta))^{1/(1-\alpha)}L$
δ	Depreciation	0.04	0.08	$r + \delta$ in capital demand

closure inside a neural operator appears to be novel, though the individual components draw on well-established methods. From an OR/MS perspective, PI-LNO instantiates the broader paradigm of AI-augmented simulation-optimization: replacing a computationally expensive forward model with a structure-preserving neural surrogate to enable estimation, sensitivity analysis, and identification diagnostics that are infeasible with the original solver.

Neural operators for parametric PDEs.. FNO [10], DeepONet [11], and their extensions [12] learn solution operators for parametric PDEs with large inference speedups over classical solvers. Physics-informed variants [13, 14] incorporate PDE residuals, improving generalization. Curriculum strategies that progressively introduce physics constraints have been explored for single-PDE settings [14], and concurrent work [15] independently proposes multi-stage curricula for FNO-class architectures; our three-stage design is specialized to the adjoint-coupled HJB-KFE system, where the order of loss activation (HJB before KFE) is dictated by the forward-backward structure. Meta-learning approaches automate loss function design for physics-informed training [16]. FNO’s spectral representation is attractive for our setting: equilibrium value functions are smooth almost everywhere, admitting compact Fourier representations with rapid spectral decay. The primary approximation challenge is the kink at the borrowing constraint, where the policy switches regime; the constrained output heads handle this boundary singularity.

Coupled multi-physics systems in scientific ML.. Neural operators have been applied to coupled PDE systems in weather and climate modeling [17], where multiple fields evolve under mutual coupling. These settings typically involve forward-in-time coupling between co-evolving fields (e.g., velocity and pressure, or magnetic and velocity fields). Our setting differs structurally: the HJB equation propagates information backward (from future utility to present value) while the KFE propagates forward (from value-implied policy to stationary distribution), and the equilibrium interest rate closes the system globally. This forward-backward coupling with endogenous closure distinguishes our physics losses from standard multi-physics residuals, though the underlying training strategy (penalizing PDE residuals alongside data loss) follows the established physics-informed paradigm.

Machine learning for economic models.. [18] train neural networks to approximate value functions for dynamic programming problems. [19] introduce deep equilibrium nets that iteratively refine approximate equilibria. [20] solve HA models with aggregate shocks using coupled neural architectures.

Their DeepHAM approach embeds separate value-function and distribution networks inside a fixed-point loop, running the economic solver during training to generate targets at each iteration. This iterative coupling limits DeepHAM to solving one model instance at a time; retraining is required for each new parameter vector. PI-LNO, by contrast, learns the full parameter-to-equilibrium mapping in a single forward pass, amortizing the cost across the entire parameter space. [21] develop neural network methods for continuous-time finance. Closest to our setting, [22] applies physics-informed neural networks to the Achdou–Bewley–Huggett continuous-time HA model, penalizing HJB and KFE residuals on a single parameter configuration; [23] uses operator learning to amortize HA solutions across parameter space. PI-LNO differs from [22] in learning the parameter-to-equilibrium operator rather than a single solution, from [23] in the adjoint-consistent discretization of the physics losses and the explicit market-clearing closure, and from both in using a Fourier backbone with FiLM conditioning. [24] approach HA models through reinforcement learning, which is complementary to our operator-learning view. The distinguishing feature of PI-LNO is that it learns the operator mapping parameters to equilibria with physics losses that match the solver’s upwind discretization, enabling rapid re-evaluation at arbitrary $\theta \in \Theta$. It neither solves a single equilibrium instance nor serves as a function approximator within a fixed-point iteration.

Surrogate-accelerated estimation.. SMM [9, 25] is the standard estimator for models without closed-form likelihoods; bootstrap inference follows [26, 27]. The computational cost of HA model estimation is well-documented [18]. [28] use neural networks to estimate nonlinear HA models with a likelihood-based approach; our contribution is complementary in targeting moment-based (SMM) estimation through a physics-informed operator surrogate, which retains explicit coupling to the continuous-time equilibrium conditions. Surrogate-accelerated inference has a growing track record in other computationally intensive fields. In cosmology, neural emulators replace Boltzmann solvers inside likelihood samplers [29]. In climate science, machine-learned parameterizations substitute for expensive subgrid-scale physics [30]. In materials science, neural network potentials trained on quantum-mechanical data accelerate molecular dynamics by orders of magnitude [31]. The common methodological pattern is replacing a costly forward model with a fast surrogate while preserving sufficient physical structure for downstream inference. Our contribution adapts this paradigm to structural economics, where the relevant “physical structure” consists of coupled optimality and distributional equilibrium conditions.

4. The PI-LNO Architecture

The PI-LNO is implemented as an FNO backbone conditioned by a learned latent code via FiLM modulation (Figure 1). It enforces three structural constraints identified in Section 1.2. First, consumption must be consistent with value-function gradients (forward-backward coupling). Second, the density must be a valid probability measure (constraint singularity). Third, the interest rate must be coupled to the density through market clearing (equilibrium closure).

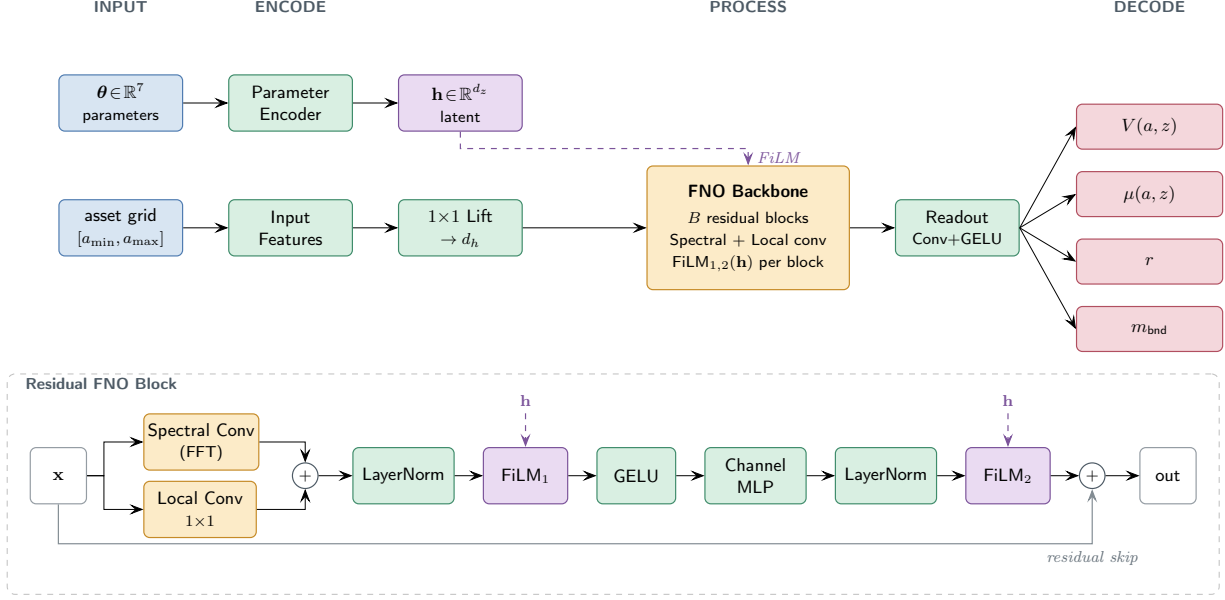


Figure 1: PI-LNO architecture. **Top:** The parameter encoder maps $\theta \in \mathbb{R}^7$ to a latent code h , which conditions the FNO backbone via FiLM modulation. The backbone processes lifted asset-grid features through B residual Fourier blocks, and the readout layer produces value functions, stationary densities, equilibrium interest rates, and boundary mass. **Bottom:** Detail of one residual FNO block showing spectral and local convolution paths, two FiLM conditioning stages, and the residual skip connection.

4.1. The Solution Operator

We define the equilibrium solution operator:

$$\mathcal{G} : \Theta \rightarrow \mathcal{X}, \quad \theta \mapsto (V^*(\cdot; \theta), c^*(\cdot; \theta), \mu^*(\cdot; \theta), r^*(\theta)), \quad (9)$$

where the value function lives in C^1 (with a kink at the constrained–unconstrained boundary), consumption in C (continuous with a possible kink), and the density in L^1_+ (nonnegative, integrates to one). The neural operator $\hat{\mathcal{G}}_\phi$ with parameters ϕ approximates this mapping.

4.2. Latent-Conditioned FNO with FiLM Modulation

The architecture has three stages: encode, process, decode.

Stage 1: Parameter encoder.. An MLP $E_\phi : \mathbb{R}^7 \rightarrow \mathbb{R}^{d_z}$ maps θ to a latent conditioning code $\mathbf{h} = E_\phi(\theta) \in \mathbb{R}^{d_z}$ with $d_z = 128$. The encoder consists of 2 hidden layers of width $d_{\text{enc}} = 256$ with GELU activations (3 linear layers total: $7 \rightarrow 256 \rightarrow 256 \rightarrow 128$). Two additional linear projections produce broadcast features: $\mathbf{p} = W_p \theta \in \mathbb{R}^{d_p}$ and $\ell = W_\ell \mathbf{h} \in \mathbb{R}^{d_\ell}$, which are replicated across the spatial grid and concatenated with the normalized asset coordinate $\tilde{a} \in [0, 1]^N$ to form the input tensor.

Stage 2: FiLM-conditioned FNO backbone.. A 1×1 convolution lifts the input to hidden dimension $d_h = 96$. The representation passes through $B = 6$ residual blocks, each combining a spectral path

with a local path:

$$\mathbf{S}_b = \mathcal{F}^{-1}[R_b \cdot \mathcal{F}[\mathbf{H}_{b-1}]_{k \leq k_{\max}}] + W_b^{\text{loc}} \mathbf{H}_{b-1}, \quad (10)$$

$$\tilde{\mathbf{S}}_b = \text{FiLM}_b^{(1)}(\text{LN}(\mathbf{S}_b), \mathbf{h}), \quad (11)$$

$$\mathbf{H}_b = \mathbf{H}_{b-1} + \text{FiLM}_b^{(2)}(\text{LN}(\text{MLP}_b(\sigma(\tilde{\mathbf{S}}_b))), \mathbf{h}), \quad (12)$$

where $R_b \in \mathbb{C}^{d_h \times d_h \times k_{\max}}$ are learnable spectral weights operating on the lowest $k_{\max} = 16$ Fourier modes, W_b^{loc} is a pointwise 1×1 convolution, and LN is channelwise layer normalization. Each block applies two FiLM modulations: one after the spectral+local path (pre-MLP), one after the MLP (pre-residual).

Each FiLM layer [32] computes $\text{FiLM}(\mathbf{x}, \mathbf{h}) = \mathbf{x} \odot (1 + \tanh(\gamma)) + \beta$ where $[\gamma, \beta] = W_{\text{film}} \mathbf{h}$. The tanh gating prevents the conditioning signal from exploding during early training when $\mathbf{h}$ is poorly calibrated. FiLM weights are initialized at small scale ($\times 0.01$) with zero bias.

Stage 3: Multi-head decoder. The backbone output $\mathbf{H}^{(B)}$ passes through a shared readout convolution ($d_h \rightarrow d_h$ with GELU) and is then decoded by four heads:

Value head. Linear projection: $\hat{V}(a_i, z_j) = \mathbf{w}_V^\top \mathbf{H}_{ij}^{(B)} + b_V$. No activation constraint is imposed, as values are unrestricted real numbers.

Policy head. Consumption is derived from value function gradients via the FOC (5): $\hat{c}(a_i, z_j) = (\partial_a \hat{V})^{-1/\sigma}$, using the same upwind finite differences employed in the physics-loss computation, rather than being predicted as an independent network output. These finite differences are a differentiable PyTorch operation executed inside the forward pass, so “single forward pass” includes the consumption computation. This construction guarantees structural consistency between the value and policy heads.

Distribution head. Linear projection to logits ℓ_{ij} , followed by elementwise softplus and normalization to a density:

$$\hat{\mu}(a_i, z_j) = \frac{\text{softplus}(\ell_{ij}) + \epsilon}{\sum_{i', j'} (\text{softplus}(\ell_{i' j'}) + \epsilon) \cdot \Delta a_{i'}}, \quad (13)$$

where Δa_i are trapezoid quadrature weights and $\epsilon = 10^{-8}$ is a density floor. This construction guarantees $\hat{\mu} > 0$ and $\sum_j \int \hat{\mu}(a, z_j) da = 1$ for any network weights ϕ .

Boundary mass head. An MLP head is available to predict the Dirac mass at $a = a_{\min}$, but is not actively supervised in the reported experiments because the current solver absorbs the borrowing-constraint mass into the first grid cell of the density rather than returning it separately (see limitations, Section 8).

Rate head. A global average pool concatenated with the latent code $\mathbf{h}$ is mapped through an MLP to $\hat{r} \in \mathbb{R}$. The MLP has two hidden layers of width 128. The predicted rate enters the HJB residual, coupling this head to the value head through the physics loss even though they are architecturally separate.

Architecture hyperparameters are summarized in Table 2. Table 2 lists the full specification used for all training runs. An earlier reduced configuration ($d_h = 48$, $d_z = 64$, $B = 4$, $k_{\max} = 12$) was

Table 2: Architecture hyperparameters used in all training runs. Total parameter count of the trained checkpoint is 1,685K (FNO) and 594K (MLP baseline); a subsequent refactoring of the spectral-weight representation in `model.py` produces a different count for the same hyperparameters.

Hyperparameter	Symbol	Value
Hidden channels	d_h	96
Latent dimension	d_z	128
Parameter broadcast	d_p	24
Latent broadcast	d_ℓ	24
FNO blocks	B	6
Fourier modes	$k_{\max}$	16
MLP expansion ratio	—	2.0
Encoder hidden dim	d_{enc}	256
Rate/boundary MLP dim	—	128

used for initial hyperparameter exploration and is retained in the codebase as an optional setting.

Motivation for the latent-conditioned design. Standard FNO spectral convolutions implicitly assume periodicity. On the bounded wealth domain $[a_{\min}, \bar{a}]$, this assumption introduces boundary artifacts precisely where the borrowing constraint singularity demands the highest accuracy. The latent conditioning addresses this limitation: FiLM modulation at every block enables the spectral layers to adapt their behavior as a function of the encoded parameters $\mathbf{h} = E(\boldsymbol{\theta})$, providing parameter-adaptive modulation without sacrificing the computational efficiency of spectral convolutions. The grid-based architecture additionally supports the normalized softplus density constraint and provides simultaneous access to all grid points for the physics residual computation.

5. Training: Preserving Structure in the Loss

The architectural constraints described in Section 4 guarantee structural validity of individual outputs; densities integrate to one and consumption satisfies the FOC. The training loss enforces a stronger condition: it penalizes outputs that violate the coupled PDE system, using residual operators that preserve the HJB–KFE adjoint duality at the discrete level.

5.1. Training Loss

The total loss combines supervised data terms, physics residuals, boundary penalties, rate-feasibility constraints, and (optionally) moment-matching terms:

$$\mathcal{L}(\phi) = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{physics}} + \mathcal{L}_{\text{boundary}} + \mathcal{L}_{\text{feasibility}} + \mathcal{L}_{\text{mom}}. \quad (14)$$

We describe each component, with particular attention to the numerical operators used for spatial derivatives.

Upwind finite differences.. Let $\Delta a_i^+ = a_{i+1} - a_i$ and $\Delta a_i^- = a_i - a_{i-1}$ denote forward and backward grid spacings (the grid is non-uniform, clustered near $a_{\min}$). The upwind derivative operator selects:

$$(D_a^{\text{uw}} f)_i = \begin{cases} (f_{i+1} - f_i)/\Delta a_i^+ & \text{if } s_i > \varepsilon_{\text{drift}} \text{ (saving),} \\ (f_i - f_{i-1})/\Delta a_i^- & \text{if } s_i < -\varepsilon_{\text{drift}} \text{ (dissaving),} \\ 0 & \text{otherwise (stationary),} \end{cases} \quad (15)$$

where s_i is the savings drift at grid point i and $\varepsilon_{\text{drift}}$ is a tolerance for the stationary regime (10^{-12} in the FD solver, 10^{-8} in the training-time physics loss for numerical stability). At boundaries: the backward difference at $i = 0$ uses the state-constraint derivative $u'(wz_j + ra_{\min})$, which corresponds to the marginal value when the household consumes all income; the forward difference at $i = N - 1$ is zero (natural upper bound). In the stationary branch ($|s_i| \leq \varepsilon_{\text{drift}}$), the derivative defaults to this same state-constraint value at the boundary.

Adjoint consistency.. The KFE is the formal adjoint of the HJB’s linear part. At the discrete level, the matrix A representing the HJB’s drift-plus-switching operator and the matrix $A^\top$ representing the KFE operator must be exact transposes. This is automatically satisfied when the same upwind convention (15) is used for both. Using different schemes (e.g., central differences for the KFE and upwind for the HJB) would break adjoint consistency and create an irreconcilable conflict in the loss.

Data losses.. Supervised fit to ground-truth equilibria from the finite-difference solver. The loss uses scale-normalized smooth L_1 (Huber) for all supervised fields (value functions, density, interest rate, consumption, boundary mass), providing robustness to outlier parameterizations:

$$\begin{aligned} \mathcal{L}_{\text{data}} = & w_V \text{SmoothL1}(V, V^*) + w_\mu \text{SmoothL1}(\mu, \mu^*) + w_r \text{SmoothL1}(r, r^*) \\ & + w_c \text{SmoothL1}(c, c^*) + w_{c,\log} \text{SmoothL1}(\log c, \log c^*) + w_b \text{SmoothL1}(m_{\text{bnd}}, m_{\text{bnd}}^*). \end{aligned} \quad (16)$$

SmoothL1 is chosen over MSE because its linear tail bounds the gradient contribution of large errors, reducing pressure on the network to sharpen $\partial_a \hat{V}$ at extreme calibrations, precisely the sharpening mechanism that causes FOC-derived consumption to overflow out of distribution (Section 6.2). MSE’s quadratic penalty would impose even stronger gradient pressure on outlier configurations, further sharpening $\partial_a \hat{V}$ and worsening OOD consumption stability. The log-space consumption loss additionally ensures that relative errors remain controlled for low-consumption households near the borrowing constraint. All terms are scale-normalized to equalize gradient magnitudes across output heads.

HJB residual.. For each predicted equilibrium, the HJB residual (4) is computed from $(\hat{V}, \hat{c}, \hat{r})$:

$$\mathcal{L}_{\text{HJB}} = \text{mean} \left[\log \left(1 + \left| \frac{\rho \hat{V}_{ij} - u(\hat{c}_{ij}) - (D_a^{\text{uw}} \hat{V})_{ij} \cdot \hat{s}_{ij} - \sum_k \Lambda_{jk} \hat{V}_{ik}}{V_{\text{scale}}} \right| \right) \right], \quad (17)$$

where the $\log(1 + |\cdot|)$ transformation softens large residuals early in training when predictions are poor, and V_{scale} is a running normalization factor.

KFE residual.. Using the adjoint-consistent conservative upwind scheme matching (6):

$$\mathcal{L}_{\text{KFE}} = \text{mean} \left[\log \left(1 + \left| \frac{F_{i+1/2,j} - F_{i-1/2,j}}{\bar{\Delta} a_i} - \sum_k \Lambda_{kj} \hat{\mu}_{ik} \right| / \mu_{\text{scale}} \right) \right], \quad (18)$$

where $F_{i+1/2,j} = \max(\hat{s}_{ij}, 0) \hat{\mu}_{ij} + \min(\hat{s}_{i+1,j}, 0) \hat{\mu}_{i+1,j}$ is the conservative flux at the cell interface.

Market-clearing loss.. The market-clearing condition (7) is enforced via:

$$\mathcal{L}_{\text{clear}} = \text{mean} \left[\log \left(1 + \left| \hat{K}^s - K^d(\hat{r}) \right| / K_{\text{scale}} \right) \right], \quad \hat{K}^s = \sum_{i,j} a_i \hat{\mu}_{ij} \Delta a_i. \quad (19)$$

Boundary and feasibility losses.. Two additional penalty terms enforce economic constraints:

$$\mathcal{L}_{\text{boundary}} = \text{SmoothL1}(\text{ReLU}(-\hat{s}_{0j}), 0) + \text{SmoothL1}(\text{ReLU}(\hat{s}_{N-1,j}), 0), \quad (20)$$

$$\mathcal{L}_{\text{feasibility}} = \text{mean} \left[\log \left(1 + \left| \text{ReLU}(r_{\min} - \hat{r}) + \text{ReLU}(\hat{r} - r_{\max}) \right| / r_{\text{scale}} \right) \right]. \quad (21)$$

The boundary loss penalizes dissaving at $a = a_{\min}$ (violates the borrowing constraint (2)) and outward drift at $a = \bar{a}$ (violates the reflecting barrier). The rate-feasibility loss penalizes interest rates outside the economically meaningful range $[r_{\min}, \rho - \varepsilon]$.

Moment loss (optional).. When enabled, we compute distributional statistics from the predicted equilibrium (wealth Gini, MPC moments, consumption dispersion, capital–output ratio; Supplementary Material, Table S1) and penalize deviations from solver-computed values via smooth L_1 . In the default configuration, moment weights are set to zero; they can be activated via a target specification file.

5.2. Training Procedure

Data generation.. We generate ground-truth equilibria using the implicit upwind solver on a non-uniform grid with $N = 64$ wealth points (power-law clustering with curvature exponent 1.5 near $a_{\min}$) and $J = 2$ income states. Optional pre-training uses $N_{\text{syn}} = 1,024$ synthetic pseudo-equilibria from calibrated closed-form approximations. The production training pipeline uses $N_{\text{sol}} = 7,500$ true solver equilibria (with 1,500 each for validation and test).

Three-stage curriculum.. The production training pipeline proceeds in three stages, each initialized from the previous stage’s checkpoint. Physics losses are introduced via a ramp schedule in Stages 2 and 3: zero weight during a warmup period (epochs 1–5), then linearly ramped from 0 to target weight over 15 epochs, then held at full weight. A checkpoint-selection guard excludes warmup epochs from best-model selection, preventing the optimizer from reverting to a checkpoint that has not yet been exposed to the physics loss. A linear learning-rate warmup over the first three

Table 3: Loss weights for solver-based fine-tuning. Left: data losses (fixed across all stages). Right: physics losses, showing the full formulation weights used in Stages 1 and 3, and the HJB-only configuration used in Stage 2.

Loss term	Weight	Physics term	Full (S3)	HJB-only (S2)
Value (smooth L_1)	1.0	HJB residual	0.5	0.1
Density (smooth L_1)	1.0	KFE residual	10^{-4}	0
Consumption (smooth L_1)	1.0	Clearing	10^{-4}	0
Consumption log (smooth L_1)	1.0	Boundary physics	0.25	0
Boundary mass (smooth L_1)	0.1	Rate feasibility	0.01	0
Interest rate (smooth L_1)	1.0			

epochs mitigates the "cold optimizer" pathology at stage transitions, where a freshly-reset AdamW optimizer has zero first-/second-moment buffers and would otherwise take oversized initial steps.

Stage 1: Supervised training on solver data (60 epochs). PI-LNO is trained from random initialization on solver-computed equilibria with all data losses active at full weight and all physics weights set to zero. The consumption loss is active from the first epoch (not tied to the physics ramp): solver-computed consumption targets are available throughout Stage 1, so there is no reason to withhold the consumption supervision signal. AdamW optimizer, learning rate 10^{-3} , weight decay 10^{-5} , cosine annealing following a three-epoch linear warmup, batch size 32, gradient clipping at $\|\cdot\|_2 \leq 0.5$.

Stage 2: HJB-only refinement (50 epochs). PI-LNO is fine-tuned from the Stage 1 checkpoint at reduced learning rate (10^{-4}) with only the HJB residual active ($w_{\text{HJB}} = 0.1$, all other physics weights zero). This stage refines the value function gradients that determine consumption via the FOC.

Stage 3: Full physics refinement (80 epochs). PI-LNO is fine-tuned from the Stage 2 checkpoint at reduced learning rate (10^{-4}) with all physics losses reactivated (full physics weights; see Table 3). This stage enforces KFE, clearing, and boundary consistency on top of the HJB-refined value function.

All stages train PI-LNO with the full architecture ($d_h = 96$, $d_z = 128$, $B = 6$, $k_{\max} = 16$) described in Table 2.

Loss weights. Table 3 lists the default loss weights used in solver-based fine-tuning.

The physics-loss weights in (14) are deliberately small relative to the data losses: the HJB residual weight (0.5) is comparable to data losses and directly shapes value-function gradients; the KFE and clearing residuals use weights of 10^{-4} (four orders of magnitude below data supervision), acting as soft regularizers that nudge solutions toward physically consistent states rather than as primary training signals. The primary training signal throughout is the supervised fit to FD-solver ground truth. In Stage 2 (HJB-only refinement), only the HJB residual is active, isolating value-function gradient refinement from density-related losses. Stage 3 uses the full physics formulation; Stage 1 uses data supervision only (all physics weights zero). Figure 2 shows validation loss and residual trajectories across all three stages.

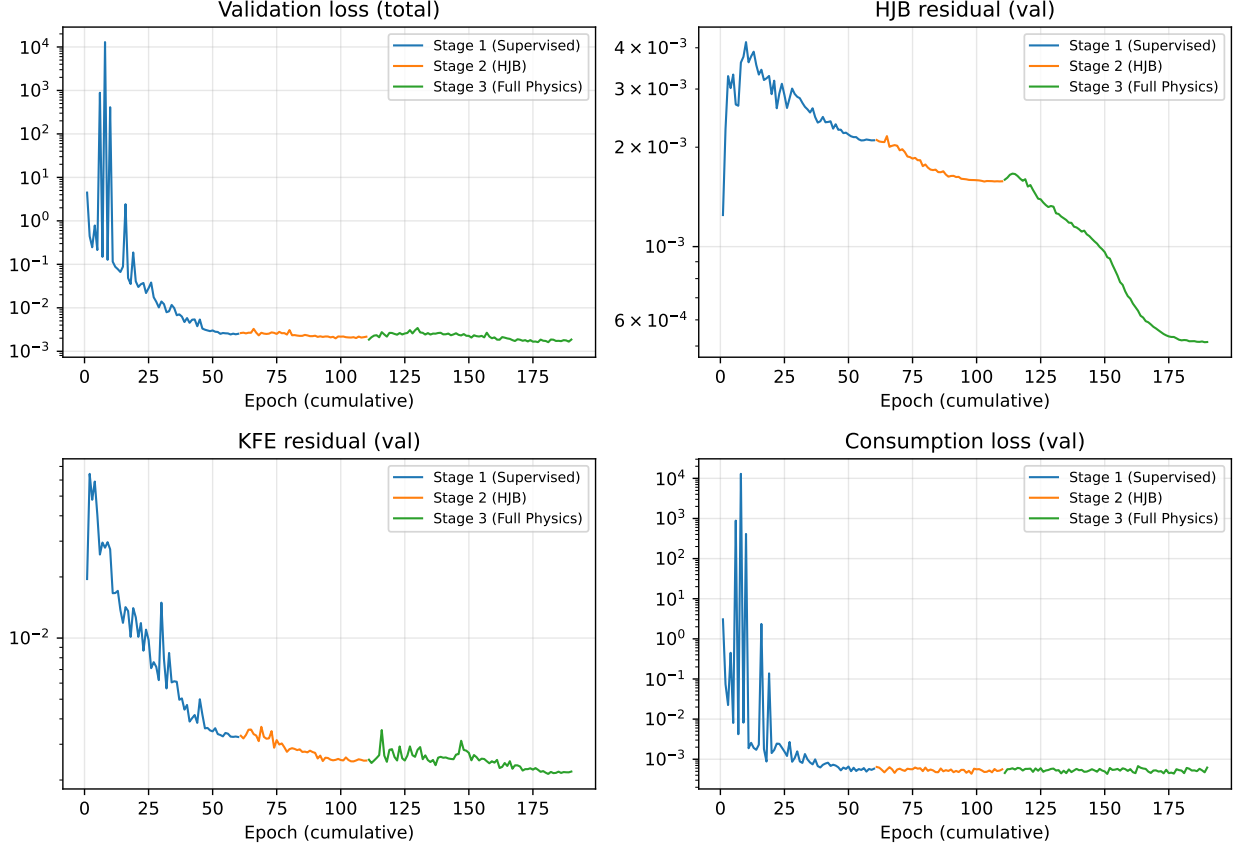


Figure 2: Training curves across all stages. Top-left: total validation loss converges in all configurations. Top-right: HJB residuals fall from ~ 0.002 in Stage 1 to ~ 0.0006 once full physics losses are active. Bottom-left: KFE residuals decrease steadily in physics-informed stages. Bottom-right: consumption loss is active at full weight from the first epoch in all stages; it remains bounded throughout the curriculum as the value function converges under supervised guidance.

6. Experiments

The structure-preservation design yields two testable predictions. First, the staged curriculum should progressively improve equilibrium accuracy as physics constraints are introduced and refined. Second, removing any structural component (adjoint consistency, the staged ramp, or constrained output heads) should produce qualitatively distinct failure modes, not merely degraded accuracy. We evaluate the first prediction via the multi-stage comparison below and the second via ablation experiments. All results are based on the two-state Poisson income model ($J = 2$) with the parameter bounds in Table 1 and the full training architecture ($d_h = 96$, $B = 6$, $k_{\max} = 16$; Table 2). Training uses 7,500 solver equilibria (1,500 validation, 1,500 test per seed) on an NVIDIA L4 GPU (24 GB, Intel Xeon CPU). Unless otherwise noted, reported accuracy numbers are means and standard deviations over five independent training seeds (42, 123, 456, 789, 1024); all seeds share the same training dataset (`DATASET_SEED = 0`), so variance across seeds reflects model initialization only, not dataset sampling. Evaluation uses the seed-specific held-out test set (1,500 equilibria).

6.1. RQ1: Equilibrium Accuracy

Table 4 reports mean absolute error (MAE) for each output field on 1,500 held-out equilibria, averaged across five independent training seeds. Stage 1 uses purely supervised training (all physics weights = 0); Stages 2 and 3 add HJB-only and full physics losses following the curriculum of Section 5.

Table 4: In-distribution accuracy (1,500 held-out equilibria, mean $\pm$ s.d. across five seeds). Normalized metrics are relative to per-field standard deviation. Bold indicates best per row.

Metric	Stage 1 (Supervised)	Stage 2 (HJB)	Stage 3 (Full)	MLP [†]
Value function (normalized)	0.018 $\pm$ 0.003	0.011 $\pm$ 0.002	0.0045 $\pm$ 0.0010	0.0063 $\pm$ 0.0003
Value, centered (normalized)	0.011 $\pm$ 0.002	0.0088 $\pm$ 0.0013	0.0045 $\pm$ 0.0010	0.0078 $\pm$ 0.0003
Density $\mu(a, z)$	0.00152 $\pm$ 0.00006	0.00116 $\pm$ 0.00003	0.00106 $\pm$ 0.00005	0.00192 $\pm$ 0.00011
Interest rate (normalized)	0.0141 $\pm$ 0.0007	0.0106 $\pm$ 0.0008	0.0089 $\pm$ 0.0012	0.0135 $\pm$ 0.0009
Consumption (normalized)	0.0104 $\pm$ 0.0011	0.0084 $\pm$ 0.0011	0.0086 $\pm$ 0.0017	0.0092 $\pm$ 0.0003
Wealth Gini	0.031 $\pm$ 0.010	0.027 $\pm$ 0.005	0.0020 $\pm$ 0.0002	0.0029 $\pm$ 0.0004
K/Y ratio MAE	0.112 $\pm$ 0.045	0.096 $\pm$ 0.020	0.012 $\pm$ 0.001	0.0164 $\pm$ 0.0013
Std(log c) MAE	0.016 $\pm$ 0.003	0.013 $\pm$ 0.002	0.0057 $\pm$ 0.0003	0.0090 $\pm$ 0.0007
Clearing residual (norm.)	0.207 $\pm$ 0.055	0.194 $\pm$ 0.027	0.082 $\pm$ 0.003	0.083 $\pm$ 0.005
HJB residual (normalized)	0.0023 $\pm$ 0.0001	0.0017 $\pm$ 0.0001	0.00059 $\pm$ 0.00010	0.00130 $\pm$ 0.00003
KFE residual (normalized)	0.0041 $\pm$ 0.0002	0.0031 $\pm$ 0.0002	0.0026 $\pm$ 0.0002	—

[†]MLP baseline: FNO blocks replaced by pointwise MLP blocks (5 seeds, single-stage training with Stage 3 physics weights and warmup/ramp schedule; 594K params vs. 1,685K for FNO).

Centered-value metric.. The raw value-function error includes a level offset arising from the CRRA utility normalization: at different risk-aversion parameters σ , the value function shifts vertically while its shape remains stable. The centered-value metric removes the per-sample, per-income-state mean before computing MAE, isolating shape error from level error. The Stage 3 model achieves 0.45% centered-value error, indicating that the equilibrium value function is captured accurately once the nuisance level shift is factored out. Figure 3 shows representative predictions overlaid on solver ground truth for four test samples; the predicted value functions and stationary densities are visually indistinguishable from the solver output.

Supervised baseline.. Stage 1 training (all physics weights zero) achieves 1.8% value-function error, 1.0% consumption error, and 3.1% wealth Gini MAE. This establishes a strong data-driven baseline and indicates that 7,500 solver equilibria provide sufficient supervision for accurate function approximation. Stability across seeds is tight: value-function s.d. is 0.3 percentage points and consumption s.d. is 0.1 percentage points, a consequence of the loss-scheduling and gradient-control choices described in Section 5.

Refinement improvements.. Adding physics losses in Stages 2 and 3 improves accuracy on most fields. Stage 3 (full physics) reduces normalized value-function error from 1.8% to 0.45%, HJB residuals from 0.0023 to 0.00059, wealth Gini MAE from 3.1% to 0.20%, K/Y ratio MAE from 0.112 to 0.012 (9 $\times$), Std(log c) MAE from 0.016 to 0.0057 ($\approx 2.8\times$), and clearing residual from 0.207 to

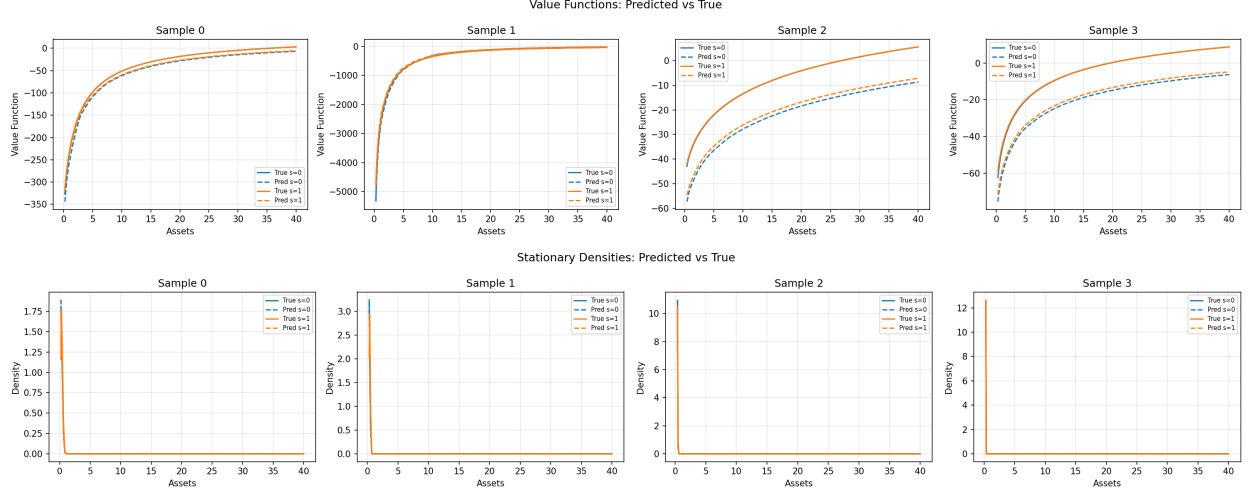


Figure 3: Predicted (dashed) vs. solver (solid) value functions and stationary densities for four held-out equilibria. The supervised-only checkpoint is shown; physics-informed stages produce similar visual agreement.

0.082 ($2.5\times$). The residual clearing normalization uses the training-set capital standard deviation (≈ 0.9 units); the 0.082 figure therefore corresponds to a mean absolute clearing gap of roughly 0.074 capital units, or about 2.5% of the mean capital stock (3.0 in the calibration). This is consistent with the observed K/Y moment MAE of 0.012 (Table 4): clearing error enters the K/Y moment in proportion to the K/Y ratio, contributing at most $0.012/3 \approx 0.4\%$ of the $\sim 3\%$ moment-surface error, with the remainder attributable to density shape and distributional approximation. The physics losses constrain the value function and its gradient simultaneously, so the improvement in the PDE residual propagates cleanly to distributional and aggregate moments. Consumption accuracy is the one exception: Stage 2 achieves the best consumption MAE (0.84%), while Stage 3 is 0.86%, slightly worse mean and 55% higher standard deviation than Stage 2, because adding full physics imposes additional gradient constraints on the value function that slightly perturb the FOC-derived policy. The Stage 3 checkpoint is recommended for applications targeting distributional and aggregate moments (K/Y , Gini, $\text{Std}(\log c)$, clearing); Stage 2 is preferable when consumption accuracy is the primary concern.

MLP baseline.. To isolate the contribution of the FNO backbone, we train a variant in which each FNO spectral block is replaced by a pointwise MLP block (same hidden dimension and depth, but no spectral mixing; 594K parameters vs. 1,685K for the FNO version due to the spectral weight tensors). The MLP baseline uses a single-stage training run with the same physics-loss weights and ramp schedule as Stage 3; it is not pre-initialized from Stages 1–2. This is not a fully matched comparison: the FNO benefits from the staged curriculum’s supervised pre-initialization, while the MLP does not. The observed gap therefore conflates architecture and training procedure to some degree; a three-stage MLP would narrow the training-schedule confound and is left for future work. What the current result establishes is that at matched hidden dimension and depth, the spectral architecture outperforms the pointwise baseline on every distributional metric. Across 5 independent seeds the

MLP reaches usable but consistently weaker accuracy: $1.4\times$ higher value-function error, $1.5\times$ higher Gini error, $1.5\times$ higher rate error, and $1.6\times$ higher $\text{Std}(\log c)$ relative to Stage 3 PI-LNO; per-seed s.d. is comparable to or smaller than the PI-LNO equivalent. The parameter-count difference also favors the FNO (1,685K vs. 594K parameters); a wider MLP could narrow the gap further.

6.2. RQ2: Out-of-Distribution Generalization

Table 5 evaluates the curriculum stages on parameters drawn from widened bounds ($\pm 10\%$ mild, $\pm 30\%$ extreme) beyond the training range (200 equilibria per level; results averaged across the same five training seeds as Table 4).

Table 5: OOD generalization at widened parameter bounds. Mean $\pm$ s.d. across five seeds. Bold indicates best per column. Consumption (norm. MAE) is shown separately; see text for the failure mode.

Checkpoint	Value (norm. MAE)		Wealth Gini MAE		Consumption (norm.)
	Mild ($\pm 10\%$)	Extreme ($\pm 30\%$)	Mild ($\pm 10\%$)	Extreme ($\pm 30\%$)	Mild / Extreme
Stage 1 (Supervised)	0.028 ± 0.003	0.275 ± 0.017	0.027 ± 0.008	0.034 ± 0.008	stable / degrades
Stage 2 (HJB)	0.019 ± 0.003	0.250 ± 0.017	0.024 ± 0.004	0.030 ± 0.006	stable / degrades
Stage 3 (Full)	0.0088 ± 0.0016	0.229 ± 0.013	0.0032 ± 0.0006	0.016 ± 0.002	see note [†] / ∞

[†] OOD consumption: both Stages 2 and 3 overflow on $\geq 4/5$ seeds at mild OOD (FOC gradient amplification; see Section 6.2); Stage 1 is most stable (3/5 seeds finite at mild OOD). All stages diverge at extreme OOD.

The staged physics curriculum provides substantial OOD regularization for value functions and distributional moments. At mild OOD ($\pm 10\%$), Stage 3 achieves normalized value-function error of 0.88% (about a $3.2\times$ improvement over the Stage 1 supervised checkpoint at 2.8%) and wealth Gini MAE of 0.0032, roughly $8.5\times$ tighter than Stage 1 (0.027). For value functions and Gini, the ordering Stage 1 $\succ$ Stage 2 $\succ$ Stage 3 holds at every OOD tier. At extreme OOD ($\pm 30\%$), all stages degrade, but Stage 3 still dominates on value and Gini: value-function error of 22.9% versus 27.5% for the supervised baseline, Gini MAE of 0.016 versus 0.034.

However, this ordering does *not* hold for consumption at OOD parameters, and the failure is not Stage 3-specific. Per-seed finite consumption counts at mild OOD are 3/5 (Stage 1), 1/5 (Stage 2), 1/5 (Stage 3); all five seeds diverge at extreme OOD across every stage. Both physics-trained checkpoints (Stages 2 and 3) overflow on ≥ 4 of 5 seeds, while supervised-only Stage 1 retains stability on a majority.

The mechanism is consistent with the structure-preservation framing: HJB training sharpens $\partial_a \hat{V}$ to satisfy the gradient-based residual, and the FOC inversion $\hat{c} = (\partial_a \hat{V})^{-1/\sigma}$ amplifies any sharpening that extrapolates outside the training box. Stage 1 leaves $\partial_a \hat{V}$ smoother because no gradient-based loss pressures it, so the FOC inversion stays bounded more often. Because the exponent $-1/\sigma$ ranges from -1 to -0.25 over the parameter box ($\sigma \in [1, 4]$), the inversion is most sensitive at low σ ; per- σ stratification of OOD failures is left to future work. For applications involving consumption moments ($\text{Std}(\log c)$, MPC) at OOD parameters, solver verification is required regardless of stage; Stage 1 is the most stable surrogate option when consumption accuracy at OOD parameters is needed.

6.3. RQ3: Ablation

Table 6 isolates the contribution of three design choices on the matched 3-stage pipeline (smooth- L_1 , LR warmup, gradient clipping throughout). All rows report 5-seed mean $\pm$ s.d. on 1,500 test equilibria.

Table 6: Ablations on the matched 3-stage pipeline (smooth- L_1 , LR warmup, gradient clipping throughout). All rows: 5-seed mean $\pm$ s.d. on 1,500 test equilibria. [†]same 3-stage curriculum as Full, with FiLM $\rightarrow$ identity. [‡]single-stage with Stage-3 physics weights. [§]MLP baseline: single-stage with Stage-3 physics weights and warmup/ramp; FNO blocks replaced by pointwise MLPs (594K params vs. 1,685K for FNO).

Configuration	Value (norm.)	Consumption (norm.)	Density MAE	Rate (norm.)
Full (Stage 3)	0.0045 $\pm$ 0.0010	0.0086 $\pm$ 0.0017	0.00106 $\pm$ 0.00005	0.0089 $\pm$ 0.0012
MLP baseline [§]	0.0063 $\pm$ 0.0003	0.0092 $\pm$ 0.0003	0.00192 $\pm$ 0.00011	0.0135 $\pm$ 0.0009
w/o FiLM [†]	0.0109 $\pm$ 0.0024	0.0111 $\pm$ 0.0015	0.00165 $\pm$ 0.00019	0.0164 $\pm$ 0.0037
w/o staged ramp [‡]	0.0057 $\pm$ 0.0014	0.0109 $\pm$ 0.0011	0.00136 $\pm$ 0.00004	0.0141 $\pm$ 0.0017

MLP baseline replaces the FNO spectral blocks with pointwise MLP blocks (same hidden dimension and depth, single-stage training with Stage 3 physics weights). The MLP is competitive on consumption and clearing but $1.8\times$ worse on density and $2.2\times$ worse on HJB residual (Table 4), consistent with the FNO’s global receptive field improving gradient-based physics losses.

Removing FiLM conditioning (3-stage curriculum with FiLM $\equiv$ identity, 5-seed) degrades value error by $2.4\times$, rate by $1.8\times$, and Gini by roughly $2.1\times$ (computed from expanded ablation runs); its role is to let the latent code modulate the spectral path so $\partial_a V$ is parameter-adapted.

Removing the staged ramp produces training instability (consumption oscillates by $10^3\times$ across consecutive epochs) that eventually recovers but at the cost of distributional accuracy: density MAE worsens by $1.3\times$ and interest-rate error by $1.6\times$ at convergence. The ramp prevents destabilizing feedback rather than merely shifting the final basin.

6.4. RQ4: Computational Speedup

Table 7 reports measured inference times on an NVIDIA L4 GPU (Intel Xeon CPU baseline), using the Stage 3 checkpoint with the full architecture (Table 2). A single forward pass takes 9.1 ms, compared to 234 ms mean for the FD solver on the same server’s CPU, a $26\times$ speedup. Batched inference exhibits superlinear scaling because the forward pass cost is dominated by fixed GPU overhead (kernel launch, synchronization) rather than per-sample computation. A single-sample call leaves most of the GPU’s cores idle; increasing the batch size to 1,000 fills the available parallelism with negligible marginal cost, since all samples share the same network weights and can execute the FFT and pointwise operations simultaneously. A 1,000-configuration batch completes in 30.8 ms (0.031 ms per sample), yielding a $7,590\times$ speedup. The FD solver cannot exploit this parallelism: each parameter configuration requires its own nested iteration (HJB convergence, KFE solve, rate bisection), with no computation shared across configurations. The comparison reflects the intended deployment scenario (a GPU surrogate replacing a CPU solver in an estimation loop) and both a GPU-accelerated FD solver and a parallelized multi-core CPU implementation would narrow the gap proportionally; the relevant claim is end-to-end estimation feasibility, not raw operator speed.

Table 7: Inference timing on an NVIDIA L4 GPU (Stage 3 checkpoint, full architecture). Solver timing from benchmark of converged configurations on the server’s Intel Xeon CPU.

Method	Total time	Per-sample	Speedup
FD solver ($\times 1$)	234 ms	234 ms	$1\times$
PI-LNO ($\times 1$)	9.1 ms	9.1 ms	$26\times$
PI-LNO ($\times 1,000$ batch)	30.8 ms	0.031 ms	$7,590\times$

The one-time training cost for a single seed (5-seed protocol) is approximately 13.6 hours across three stages (4.3 h supervised, 3.6 h HJB-only, 5.7 h full physics) on a TPU v6e-8. At the single-query speedup of $26\times$, the surrogate recoups this cost after roughly $48,960\text{ s}/(0.234 - 0.009)\text{ s} \approx 218,000$ solver-equivalent evaluations; in batched deployment ($7,590\times$) the break-even is similar since the dominant cost is training, not inference. A typical SMM estimation run involves 10^3 – 10^5 evaluations per parameter vector, so the surrogate pays for itself within one estimation campaign.

7. Application: Structural Estimation

This section demonstrates the feasibility of surrogate-accelerated structural estimation using PI-LNO. The goal is not to report final empirical estimates (the two-state income model is too stylized for that purpose; Section 8) but to show that the surrogate is accurate enough to make SMM estimation mechanically feasible and to illustrate diagnostics that become practical only with a fast surrogate.

7.1. Simulated Method of Moments

Given empirical moment targets $\bar{\mathbf{m}} \in \mathbb{R}^M$ and moment weights $w_k > 0$, the SMM estimator [9, 25] solves:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta} \in \Theta} \sum_{k=1}^M w_k \left(\frac{m_k(\hat{\mathcal{G}}_{\phi}(\boldsymbol{\theta})) - \bar{m}_k}{\bar{m}_k} \right)^2, \quad (22)$$

where $\mathbf{m}(\cdot)$ extracts moments from predicted equilibrium objects. The objective uses relative errors so that moments of different scales contribute comparably. Each evaluation requires only a single forward pass through the operator. We employ multi-start L-BFGS-B (SciPy) with numerical gradients, using multiple random initializations to mitigate sensitivity to local optima. The number of restarts is configurable; the default configuration uses five restarts.

7.2. Empirical Targets

We target five moments selected for economic content and data availability (Table 8). The diagonal weighting matrix reflects independent data sources: the Survey of Consumer Finances for wealth, Norwegian lottery data for MPCs, the Panel Study of Income Dynamics for consumption, and national accounts for capital–output ratios. MPC targets from [33] derive from Norwegian administrative data; we adopt them as proxies in the absence of comparable US estimates.

Table 8: Empirical moment targets for structural estimation.

Moment	Source	Target	SE
Wealth Gini	SCF [34]	0.81	0.015
Mean MPC	[33]	0.25	0.04
90th pctl MPC	[33]	0.55	0.08
Std(log c)	[35]	0.22	0.03
K/Y ratio	BEA NIPA	3.0	0.20

7.3. Identification Diagnostics

We compute the Jacobian $J = \partial \mathbf{m} / \partial \boldsymbol{\theta} \in \mathbb{R}^{M \times 7}$ via central finite differences, normalize to elasticities $E_{ij} = (\partial m_i / \partial \theta_j)(\theta_j / m_i)$, and take the SVD $E = U \Sigma V^\top$ [36]. The condition number $\kappa = \sigma_{\max} / \sigma_{\min}$ measures identification strength; right singular vectors in V reveal which parameter directions are most informed by the data. This analysis is rarely performed for HA models because it requires $2 \times 7 = 14$ additional solver evaluations. With the PI-LNO surrogate, the marginal cost of these evaluations is negligible (< 1 second total), and the results are reported jointly with the synthetic recovery in Section 7.5.

7.4. Parametric Bootstrap

The surrogate’s speed naturally enables parametric bootstrap inference [26, 27]: for each draw $b = 1, \dots, B$, sample perturbed targets $\bar{\mathbf{m}}^{(b)} \sim \mathcal{N}(\bar{\mathbf{m}}, \text{diag}(\text{SE}^2))$, re-estimate $\hat{\boldsymbol{\theta}}^{(b)}$ via (22), and construct percentile CIs from $\{\hat{\boldsymbol{\theta}}^{(b)}\}_{b=1}^B$. This procedure is computationally feasible only with a fast surrogate; $B = 200$ draws with multiple restarts would require $\mathcal{O}(10^6)$ solver evaluations but only minutes with PI-LNO. Implementation is deferred to future work.

7.5. Synthetic Recovery

The standard validation for surrogate-based estimation is synthetic recovery: generate moment targets from the exact solver at a known $\boldsymbol{\theta}^*$, recover $\hat{\boldsymbol{\theta}}$ using the surrogate, and report the bias $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$. This is distinct from matching empirical targets, which tests whether the model is well-specified; synthetic recovery tests whether the *surrogate* is reliable conditional on the model being correct.

We evaluate three interior parameter configurations (Table 9): point A is a central calibration; point B has low risk aversion ($\sigma = 1.5$) and loose borrowing constraint; point C has high risk aversion ($\sigma = 3.0$) and a tighter constraint. For each point, the exact solver generates ground-truth moments; the surrogate SMM (differential evolution, 300 iterations) recovers $\hat{\boldsymbol{\theta}}$.

The key finding is in Tables 9–10: the surrogate matches its own target moments to near-machine precision (SMM objectives 2.8×10^{-7} , 2.3×10^{-8} , 5.0×10^{-6}), yet the estimated parameters are far from $\boldsymbol{\theta}^*$ on σ and $\bar{z}$. A control test using the FD solver as the moment function (“Control: FD-SMM” below) shows $\boldsymbol{\theta}^*$ is a well-separated basin (5 starts, 7D space), but identification is *weak* along the $(\sigma, \bar{z})$ direction (surrogate $\kappa \approx 30$, FD-solver $\kappa \approx 15$ at Point A; Table 10). A small ($\sim 3\%$) surrogate moment-surface error then amplifies into a large $(\sigma, \bar{z})$ parameter bias along that ridge.

Table 9: Synthetic recovery bias $\hat{\theta} - \theta^*$ (seed 42 Stage 3, differential evolution, 300 iterations). SMM objectives at convergence: A = 2.8×10^{-7} , B = 2.3×10^{-8} , C = 5.0×10^{-6} ; max relative moment error 0.04% (A), 0.01% (B), 0.19% (C). Large $\sigma/\bar{z}$ biases reflect weak global identification along the $(\sigma, \bar{z})$ direction (Table 10) interacting with $\sim 3\%$ surrogate moment-surface error; the FD-SMM control shows θ^* is a well-separated basin (5 starts, 7D space).

Parameter	Point A (central)		Point B (low σ)		Point C (high σ)	
	θ^*	Bias	θ^*	Bias	θ^*	Bias
ρ (discount)	0.050	+0.006	0.040	+0.013	0.060	+0.017
σ (risk aversion)	2.0	+0.966	1.5	+1.568	3.0	−0.844
$a_{\min}$ (borrowing)	0.10	+0.016	0.05	−0.009	0.20	+0.007
λ (separation)	0.15	−0.009	0.10	−0.002	0.20	+0.056
$\bar{z}$ (productivity)	1.00	+0.175	1.20	+0.417	1.00	+0.024
α (capital share)	0.33	−0.002	0.30	+0.060	0.35	+0.017
δ (depreciation)	0.060	+0.005	0.050	+0.015	0.070	−0.023

Identification diagnostics. To diagnose the synthetic recovery results, we compute the elasticity Jacobian $E_{ij} = (\partial m_i / \partial \theta_j)(\theta_j / m_i)$ at θ^* via central differences using both the surrogate ($2 \times 7 = 14$ surrogate evaluations, < 1 second) and the exact FD solver (14 solver evaluations, < 1 second), and take the SVD of each. Table 10 reports the results.

Table 10: SVD identification diagnostics at each true θ^* . σ_1 = largest singular value; σ_5 = smallest; κ_{surr} = surrogate condition number; κ_{FD} = FD-solver condition number (same 14 finite-difference evaluations, < 1 second). Weakest direction = largest absolute loading in the last right singular vector of the surrogate elasticity Jacobian.

Point	σ_1	σ_5	κ_{surr}	κ_{FD}	Weakest direction
A (central)	4.25	0.141	30	15	ρ (loading +0.87)
B (low σ)	2.62	0.096	27	16	δ (loading −0.66)
C (high σ)	7.74	0.093	84	190	λ (loading −0.80)

With $M = 5$ moments and $K = 7$ parameters, the elasticity Jacobian $E \in \mathbb{R}^{5 \times 7}$ has rank at most 5. The two zero singular values define a two-dimensional first-order null space: no linear combination of moments responds, at first order, to perturbations in those directions. At Point A, the null directions are v_6 (dominated by λ , loading +0.95) and v_7 (dominated by $\bar{z}$, α , δ , with $\bar{z}$ loading +0.68); identification in these directions relies entirely on moment curvature. This is categorically different from weak identification (small but positive σ_5): a small σ_5 means moments barely respond; the null directions mean they do not respond at first order at all.

Surrogate condition numbers of 27–84 and FD-solver condition numbers of 15–190 both confirm weak identification, and the two Jacobians agree qualitatively. The FD values are lower than the surrogate at Points A and B ($\kappa_{\text{FD}} \approx 15$ –16 vs $\kappa_{\text{surr}} \approx 27$ –30) and substantially higher at Point C ($\kappa_{\text{FD}} = 190$ vs $\kappa_{\text{surr}} = 84$), indicating the surrogate Jacobian is a reliable qualitative guide to identification structure but not a precise quantitative one. The smallest singular value (≈ 0.09 –0.14) is consistent across all three points and both solvers.

σ does not appear as the weakest local direction in any of the SVDs, yet has the largest recovery biases. Projecting the mean seed-42 bias direction ($\Delta\sigma = +0.74$, $\Delta\bar{z} = +0.16$) onto the surrogate right singular vectors at Point A reveals that 95% of the bias loads on v_1 (the strongest identified

direction, $\sigma_1 = 4.25$), with only 15% on the null direction v_7 . This means the σ bias is not primarily a weak-identification effect; it is surrogate approximation error in a direction the moments can distinguish, amplified by the $(\sigma, \bar{z})$ moment-surface shape. The $\bar{z}$ component of the bias has a partial null-space contribution (v_7), consistent with first-order non-identification of $\bar{z}$ in isolation. The FD-SMM control (see below) confirms that curvature-based identification succeeds globally at all three points, establishing that the system is identified at second order even where linear identification fails.

Multi-seed recovery at Point A.. Re-running the recovery against each of the 5 trained checkpoints (independent training seeds, same Point A target) gives SMM objective $< 1.7 \times 10^{-6}$ in all cases. Bias means $\pm$ s.d. across the 5 seeds: $\sigma = +0.74 \pm 0.59$, $\bar{z} = +0.16 \pm 0.06$, $\rho = +0.008 \pm 0.008$, $\lambda = -0.022 \pm 0.016$, $a_{\min} = +0.012 \pm 0.002$, $\alpha = +0.003 \pm 0.018$, $\delta = +0.001 \pm 0.005$. Non-collinear parameters ($\rho, a_{\min}, \lambda, \alpha, \delta$) are recovered consistently across seeds with bias s.d. under 0.02 because they are well-identified and unaffected by the same shape error.

Control: FD-SMM at all three points.. To separate surrogate-shape error from true non-identification, we re-ran SMM at all three calibration points using the exact FD solver as the moment function, with multi-start L-BFGS-B from θ^* and four perturbations (roughly 4,000 solver evaluations, 4 minutes per point). At all three points, the optimizer started at θ^* converges to θ^* exactly with loss 0. Perturbed starts reach local minima with substantially higher loss: 4×10^{-4} – 5×10^{-2} at Point A; 10^{-2} – 10^{-1} at Point B; roughly 2×10^{-2} at Point C (three of four perturbed starts hit solver singularities and return the 10^{12} penalty). All non-degenerate local traps are substantially worse than the global optimum at loss 0, confirming they are local traps rather than near-equivalent optima. Perturbed starts at B and C exhibit $|\Delta\sigma| \leq 0.57$ and $|\Delta\bar{z}| \leq 0.14$, with σ and $\bar{z}$ moving in opposing directions consistent with the $(\sigma, \bar{z})$ identification ridge. Independently, the surrogate’s recovered $\hat{\theta}$ from each seed produces FD-evaluated moments off by 2–5% relative. **Interpretation:** with five starts in a seven-dimensional space these results cannot rule out other global optima, but every perturbed start converges to a substantially worse local trap, making θ^* a well-separated basin. Identification is *weak* along the $(\sigma, \bar{z})$ direction ($\kappa_{\text{surr}} \approx 30$, $\kappa_{\text{FD}} \approx 15$ at Point A; Table 10). A small ($\sim 3\%$) surrogate moment-surface error therefore amplifies into a large $(\sigma, \bar{z})$ parameter bias, primarily via approximation errors in the strongly-identified v_1 direction. The seed-to-seed s.d. in σ reflects different surrogates’ shape errors interacting with the same weakly-identified ridge.

7.6. Empirical Fit

The two-state Aiyagari economy cannot simultaneously match all five empirical targets. The best fit is achieved for K/Y and $\text{Std}(\log c)$; larger discrepancies arise for the wealth Gini and MPC moments. The two-state income process cannot generate the extreme right tail of the wealth distribution [34], and the single-asset structure limits MPC heterogeneity. These discrepancies reflect model limitations, not surrogate error: the finite-difference solver produces the same moment mismatch at the estimated parameters.

Surrogate bias.. The operator introduces approximation error that propagates into parameter estimates. For publication-quality estimates, finite-difference refinement at the converged $\hat{\theta}$ is recommended: re-solve the equilibrium with the exact solver at the surrogate’s optimum and verify that moments are unchanged within tolerance. Extending to a richer income process (3+ states or continuous) and performing FD refinement are natural next steps for a full empirical application.

8. Discussion

The structure-preservation principle provides a principled basis for assessing when the surrogate can and cannot be trusted.

8.1. Applicability Guidelines

PI-LNO is best suited to settings requiring hundreds to thousands of equilibrium evaluations (estimation, counterfactual analysis, sensitivity studies) where parameters lie strictly within the training box (Table 1) and moment-level accuracy suffices. The finite-difference solver remains preferable when few evaluations are needed, parameters approach or exceed the training bounds, or the application demands accuracy below PI-LNO’s achievable tolerance. Users should treat the training bounds as hard limits rather than soft guidelines: error growth outside the box is not gradual and predictable but depends on the local geometry of the equilibrium manifold in ways that are difficult to anticipate without solver verification. Hybrid approaches, in which the operator warm-starts the iterative solver or provides coarse exploration with subsequent solver refinement, offer a robust intermediate alternative that combines surrogate speed with solver accuracy guarantees.

8.2. Limitations

Two-state income restriction.. The physics losses are structurally coupled to the two-state Poisson income process: the HJB residual and the adjoint-consistent KFE residual both depend on the specific sparsity pattern of the 2×2 generator matrix Λ . Changing the number of income states is not a hyperparameter change; it requires rewriting the discrete operators, retraining from scratch on a new dataset, and verifying adjoint consistency for the new generator structure. The FNO backbone and FiLM conditioning are agnostic to the number of income states, but the physics loss layer is not, so the two-state model should not be applied to economies with richer income heterogeneity.

Out-of-distribution degradation.. PI-LNO is an interpolator over the seven-dimensional parameter box in Table 1. Error grows monotonically as parameters move away from the training distribution: within the box the model achieves the accuracy reported in Table 4, but accuracy degrades as parameters approach and cross the boundary, and extrapolation beyond the box is unreliable by construction. This is not a failure of the surrogate design; it is an intrinsic property of any function approximator trained on a bounded domain, but it must be taken seriously in applied work. In particular, the OOD degradation interacts with the $J=2$ structural constraint: the seven parameters do not include the number of income states or the full generator matrix, so parameter combinations that implicitly require different qualitative income dynamics (e.g., near-degenerate switching, very

high persistence) will produce large errors even within the nominal training box. Applied users should verify that their target parameter configuration lies well within the training bounds and cross-check at least a few evaluation points against the exact solver before using the surrogate for inference.

Borrowing constraint mass point.. The distribution μ^* has a true Dirac component at $a = a_{\min}$. Our architecture includes a boundary mass head designed to capture this atom, but the current solver absorbs the mass point into the first grid cell of the density rather than reporting it separately. Consequently, the boundary mass target is zero in the reported experiments and the head is not actively supervised. Connecting the head to a solver that extracts an explicit mass point, or to models with multiple binding constraints, requires additional implementation.

Aggregate shocks.. We solve for stationary equilibria only. Extending to models with aggregate risk requires the operator to map $(\theta, \text{aggregate state}) \rightarrow \text{equilibrium}$, increasing input dimensionality and requiring time-dependent solutions.

Model misspecification.. The two-state economy cannot match the full empirical wealth distribution, which requires persistent income, bequests, or entrepreneurial risk. It also cannot reproduce the degree of MPC heterogeneity observed in the data, which requires richer asset structure. The estimation pipeline correctly reveals these limitations through large moment fit errors.

Surrogate bias in estimation.. See the surrogate bias discussion in Section 7.

Grid resolution.. The solver uses $N = 64$ asset grid points, which is standard in the Achdou et al. [4] framework and sufficient for smooth solutions. Formal grid convergence tests against finer grids ($N = 128, 256$) have not been conducted; users working with parameterizations that generate sharp density spikes near the borrowing constraint should verify solver accuracy at their target resolution before relying on the surrogate.

Open directions.. The most immediate extension is multi-state income ($J \geq 3$): the FNO backbone and FiLM conditioning are agnostic to J , but the physics loss layer must be rederived for the new generator sparsity pattern, requiring a fresh training run on a J -specific dataset. The current architecture ($d_h = 96, B = 6$) was selected by preliminary tuning; systematic ablations across model sizes, Fourier-mode counts, and loss balancing remain open. Combining the staged curriculum with higher-order derivative regularization, or with adaptive loss weighting schemes, could further tighten gradient-derived quantities.

9. Conclusion

The central contribution of this paper is to demonstrate that *structure preservation* (not merely pointwise accuracy) determines whether a neural surrogate is suitable for economic inference. We develop this principle in the context of the continuous-time Bewley–Huggett–Aiyagari economy with

two-state Poisson income, where preserving structure imposes three concrete requirements. First, the discrete HJB and KFE residuals in the loss must be exact adjoints of each other, maintaining consistency between the forward and backward equations. Second, consumption must be derived from value-function gradients via the FOC rather than predicted independently. Third, physics losses must be introduced via a staged curriculum, after the supervised fit has stabilized, to circumvent the inherent instability of simultaneous forward-backward optimization. Each design choice addresses a specific pathology of the coupled equilibrium system; our ablation experiments on a matched 5-seed protocol confirm that FiLM conditioning contributes $\sim 2.4\times$ on value error and $\sim 2\times$ on distributional moments, and that removing the staged ramp degrades distributional accuracy. An MLP baseline (FNO blocks replaced by pointwise MLPs, single-stage training, 594K vs. 1,685K parameters) is consistently weaker on every distributional metric; the gap conflates architecture and training differences, but the spectral architecture is superior at matched hidden dimension and depth (Section 6.3). The resulting PI-LNO surrogate produces outputs that are structurally coherent: they approximately satisfy the same system of equations that defines the true equilibrium. Beyond accuracy, the surrogate’s speed enables dense identification diagnostics that are impractical with the FD solver alone; FD-SMM controls at all three evaluation points show θ^* is a well-separated basin (5 starts, 7D space), with the parameter-recovery bias attributable to weak global identification interacting with small surrogate moment-surface error.

Natural extensions include multi-state income processes, aggregate shocks, and multi-asset economies, which would broaden the applicability of the approach to the full range of heterogeneous-agent models employed in quantitative macroeconomics. Combining the surrogate with automatic differentiation for gradient-based estimation and with active learning for adaptive training-set design constitute particularly promising directions for future research.

More broadly, the structure-preservation principle demonstrated here (enforcing adjoint consistency, constraining output heads, and staging physics losses) applies to any simulation-based optimization setting where the forward model involves coupled multi-physics PDEs with endogenous closure conditions. Settings with a similar structure (costly forward models, endogenous closure conditions, and simulation-based inference) arise in operations research (e.g., supply chain network equilibria [37], traffic assignment models [38]) and could benefit from the same surrogate-accelerated methodology, provided the physics losses are adapted to the relevant equilibrium conditions.

Code.. Code will be released upon publication.

Data availability

All data used in this study are synthetically generated by the finite-difference equilibrium solver described in the Supplementary Material. No empirical datasets were used. Code, trained model weights, and solver-generated training data are available at: <https://zenodo.org/doi/10.5281/zenodo.XXXXXXX> (link will be activated upon acceptance; a private reviewer link is available from the corresponding author on request).

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CRedit authorship contribution statement

Brandon Yee: Conceptualization, Methodology, Supervision, Visualization, Writing – review & editing. **Anqi Peter Li:** Software, Formal analysis, Investigation, Data curation, Resources, Writing – original draft. **Pairie Koh:** Supervision, Writing – review & editing.

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S1. Equilibrium Moments

All integrals are evaluated using the trapezoidal rule. The local marginal propensity to consume is defined as $\text{MPC}(a, z) = \partial c^* / \partial a$, computed via finite differences and capped at 5.0 to prevent outlier contamination.

Table S11: Equilibrium moments computed from predicted output.

Moment	Definition
Mean assets	$\mathbb{E}[a] = \sum_z \int a \mu(a, z) da$
Wealth Gini	$1 - \sum_i w_i (L_i + L_{i-1})$ (Lorenz curve)
Top 10% / 1% wealth share	$\sum_{a \geq a_p} w_i a_i / \mathbb{E}[a]$
Share at borrowing limit	$\sum_z \mu(a_{\min}, z) + m_{\text{bnd}}(z)$
Mean consumption	$\sum_z \int c^* \mu da$
Std / Std log consumption	weighted std of $c / \log c$ over μ
Mean local MPC	$\sum_z \int \text{MPC}(a, z) \mu da$
90th pctl local MPC	MPC at 90th percentile of μ
K/Y ratio	$K / (AK^\alpha L^{1-\alpha})$

S2. Solver Algorithm

The solver employs three nested iterative procedures:

1. **Interest rate bisection.** Bracket $[r_{\min}, r_{\max}]$ via 24-point grid search. Iterate: solve HJB at r , compute KFE density, evaluate $K^s(r) - K^d(r)$, bisect until $|K^s - K^d| < 10^{-5}$ (up to 120 iterations).
2. **HJB iteration.** Implicit scheme: $(I + \Delta t A) V^{(n+1)} = V^{(n)} + \Delta t u(c^{(n)})$, where A is the tridiagonal upwind operator and $\Delta t = 1000$. Converge when $\|V^{(n+1)} - V^{(n)}\|_\infty < 10^{-7}$ (up to 400 iterations).
3. **KFE solve.** Given converged $s^*(a, z)$, solve $A^\top \mu = 0$ with $\sum \mu \Delta a = 1$.